

5. Carathéodory Measurability and the Coincidence Criterion for Measures

REAL VARIABLES I

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The preceding notes defined measures on σ -algebras, while an outer measure is defined on every subset of a set X . According to Vitali's counterexample, an outer measure need not be additive even on two disjoint sets. In this section we identify a criterion that allows us, for any outer measure μ to be able to find σ -algebra contained in 2^X on which μ is actually a measure (it is additive). For Lebesgue outer measure, the resulting σ algebra contains every Borel set.

Finally, we prove a uniqueness principle: two measures that coincide on a suitable generating family coincide on the generated σ -algebra. This will allow us to prove that on the Borels, the Lebesgue measure is indeed the unique one that assigns to closed intervals their lengths.

After studying these notes, you will be able to do the following.

- Apply the Carathéodory criterion and prove closure properties of the measurable sets it defines.
- Construct a complete measure by restricting an outer measure to its Carathéodory measurable sets.
- Relate Lebesgue measurable sets to Borel sets and measure completion.
- Use Dynkin's π - λ theorem to prove that finite or σ -finite measures coincide on a generated σ -algebra.

5.1 The Carathéodory criterion

Throughout this section, X is a set and $\mu^*: \mathcal{P}(X) \rightarrow [0, \infty]$ is an outer measure. Thus $\mu^*(\emptyset) = 0$, μ^* is monotone, and σ subadditive.

The central idea for defining the set A to be measurable is to require that one can "cut" any set $E \subset X$. Using A . More precisely, given any $E \subset X$ it holds $E = (E \cap A) \cup (E \cap A^c)$ and these two sets are disjoint.

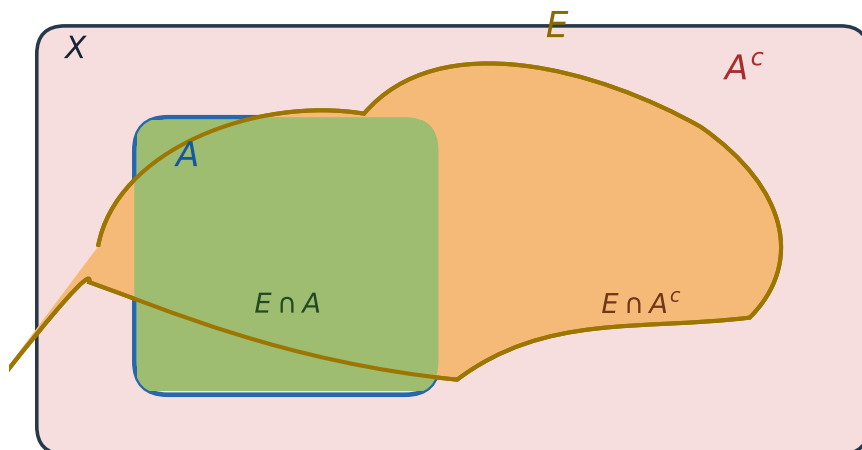


Figure 1: Within the universe X , blue represents A , red represents A^c , and yellow represents E . Their overlap is green in $E \cap A$ and orange in $E \cap A^c$.

Figure description: Inside one large rounded rectangle labelled X , a blue rounded rectangle labelled A leaves space around it in red A complement. A yellow irregular set E crosses the boundary of A . The colors blend to green in E intersection A and orange in E intersection A complement.

Subadditivity of μ^* always gives $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \setminus A)$. For measurability we require the reverse inequality.

Definition 5.1.1 (Carathéodory measurable set). A set $A \subseteq X$ is *Carathéodory measurable with respect to μ^** if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \setminus A) \quad \text{for every } E \subseteq X. \quad (1)$$

The collection of all such sets is denoted by $\mathcal{M}_{\mu^*}$.

Note that the above definition imposes no restriction on the set $E \subseteq X$ being cut. It is immediate to check that $\emptyset$ and X are always Carathéodory measurable.

Let $\mathcal{C}_{\mu^*}$ be the set of Carathéodory measurable sets for μ^* . Our goal is to show that:

- The outer measure $\mathcal{C}_{\mu^*}$ is finitely additive on disjoint sets in $\mathcal{C}_{\mu^*}$;
- $\mathcal{C}_{\mu^*}$ is a σ algebra.

That way, restricting μ^* to $\mathcal{C}_{\mu^*}$ we obtain a measure on a σ -algebra.

Proposition 5.1.2 (Finite additivity with a measurable union). Let $A, B, A \cup B \in \mathcal{M}_{\mu^*}$. If $A \cap B = \emptyset$, then

$$\mu^*(A \cup B) = \mu^*(A) + \mu^*(B).$$

Proof. (proves 5.1.2)

Apply the Carathéodory criterion for A to the test set $A \cup B$. Since $A \cap B = \emptyset$, it gives

$$\mu^*(A \cup B) = \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \setminus A) = \mu^*(A) + \mu^*(B).$$

□

We next prove the closure properties of the Carathéodory domain.

Proposition 5.1.3 (Finite unions and complements). The collection $\mathcal{M}_{\mu^*}$ is closed under finite unions and complements. Consequently, it is also closed under finite intersections and set differences.

Proof. (proves 5.1.3)

Step 1: prove closure under finite unions.

Let $A, B \in \mathcal{M}_{\mu^*}$ and fix $E \subseteq X$. The diagram in [Figure 2](#) shows the four pieces obtained by making both cuts.

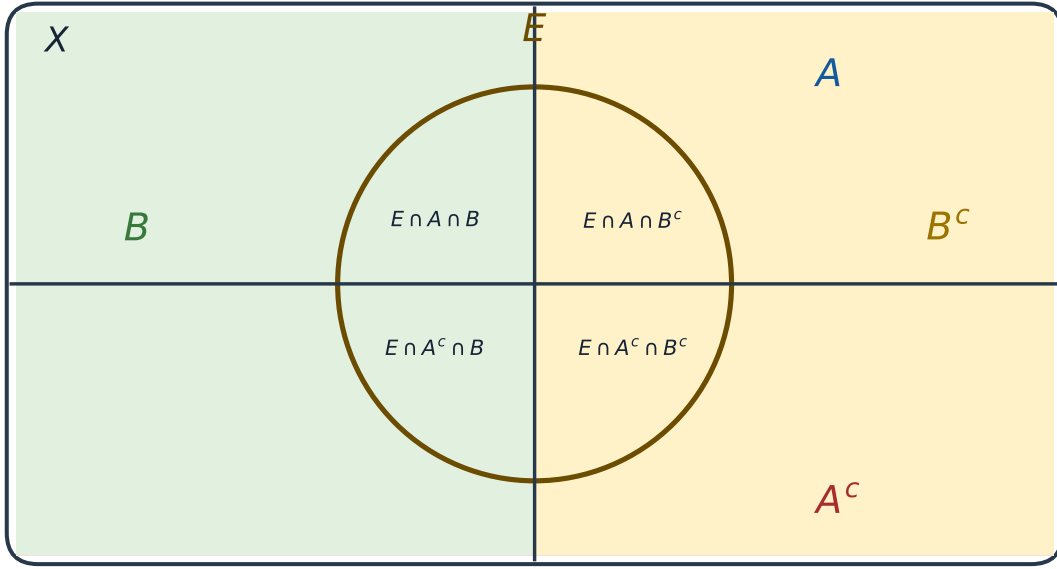


Figure 2: The horizontal cut by A and A^c , followed by the vertical cut by B and B^c , divides E into four pieces.

Figure description: Inside a rectangle labelled X , a horizontal line divides A above from A complement below, and a vertical line divides B on the left from B complement on the right. A central circle labelled E is quartered and its four pieces are labelled E intersection A intersection B , E intersection A intersection B complement, E intersection A complement intersection B , and E intersection A complement intersection B complement.

The two cuts give

$$\begin{aligned} \mu^*(E) &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &\geq \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c). \end{aligned}$$

The first inequality holds by measurability of A . The second holds by measurability of B , applied to $E \cap A$ and $E \cap A^c$, respectively. Since the first three pieces cover $E \cap (A \cup B)$, subadditivity gives

$$\begin{aligned} \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) \\ \geq \mu^*(E \cap (A \cup B)). \end{aligned}$$

The fourth piece is $E \cap (A \cup B)^c$. Combining this inequality with the preceding lower bound gives

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c).$$

The reverse inequality follows from subadditivity, so $A \cup B$ satisfies the Carathéodory criterion. Induction gives closure under finite unions.

Step 2: prove closure under complements.

The defining equality is symmetric in A and A^c , because $E \cap A^c$ is the part of E outside A . Thus $A \in \mathcal{M}_{\mu^*}$ implies $A^c \in \mathcal{M}_{\mu^*}$.

Step 3: recover intersections and differences.

De Morgan's laws give closure under finite intersections, and $A \cap B^c$ gives closure under differences. $\square$

Proposition 5.1.4 (Countable unions of measurable sets). The collection $\mathcal{M}_{\mu^*}$ is closed under countable unions. Consequently, it is a σ -algebra.

Proof. (proves 5.1.4)

Step 1: treat pairwise disjoint unions.

Let $(A_n \in \mathcal{M}_{\mu^*})_{n \geq 1}$ be pairwise disjoint and set

$$A = \bigcup_{n=1}^{\infty} A_n.$$

Fix $E \subseteq X$. Repeated application of the criterion gives, for every $N \geq 1$,

$$\mu^*(E) = \sum_{n=1}^N \mu^*(E \cap A_n) + \mu^*\left(E \setminus \bigcup_{n=1}^N A_n\right).$$

By monotonicity,

$$\mu^*(E) \geq \sum_{n=1}^N \mu^*(E \cap A_n) + \mu^*(E \setminus A).$$

Taking the supremum over N , and then applying countable subadditivity to $E \cap A = \bigcup_{n=1}^{\infty} (E \cap A_n)$, gives

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \setminus A).$$

The reverse inequality follows from subadditivity. Thus $A \in \mathcal{M}_{\mu^*}$.

Step 2: disjointify an arbitrary countable union.

Let $(B_n \in \mathcal{M}_{\mu^*})_{n \geq 1}$ be arbitrary. Define

$$D_1 = B_1, \quad D_n = B_n \setminus \bigcup_{k=1}^{n-1} B_k \quad (n \geq 2).$$

By Proposition 5.1.3, every D_n is measurable. The sets D_n are pairwise disjoint and have the same union as the sets B_n , so their union is measurable by Step 1. Since $X \in \mathcal{M}_{\mu^*}$ and the collection is closed under complements, it is a σ -algebra. $\square$

5.1.1 Completeness of measures

Let $\mathcal{A}$ be a σ -algebra on X , and let ν be a measure on $\mathcal{A}$. We say that $\mathcal{A}$ is *complete with respect to ν* if for any $N \in \mathcal{A}$ with $\nu(N) = 0$ and any $M \subset N$ one has $M \in \mathcal{A}$. An equivalent characterization is that if $A \in \mathcal{A}$ and $B \subset X$ such that $A \Delta B \subset N$ for some $N \in \mathcal{A}$ with $\nu(N) = 0$ then $B \in \mathcal{A}$.

Corollary 5.1.5 (Carathéodory construction). The restriction $\mu = \mu^*|_{\mathcal{M}_{\mu^*}} : \mathcal{M}_{\mu^*} \rightarrow [0, \infty]$ is a measure.

The preceding propositions establish the required facts. They show that $\mathcal{M}_{\mu^*}$ is a σ -algebra and that μ^* is finitely additive on its pairwise disjoint measurable sets. The result from the previous notes therefore gives σ -superadditivity on pairwise disjoint families. Together with the σ -subadditivity of the outer measure, this gives σ -additivity of the restriction.

Proposition 5.1.6 (Completeness of the Carathéodory σ -algebra). The σ -algebra $\mathcal{M}_{\mu^*}$ is complete with respect to the measure $\mu = \mu^*|_{\mathcal{M}_{\mu^*}}$.

Proof. (proves 5.1.6)
Exercise. □

Example 5.1.7 (Weighted counting measure). Let $w: X \rightarrow [0, \infty]$ and define

$$\mu_w^*(E) = \sum_{x \in E} w(x), \quad E \subseteq X,$$

where the sum is the supremum of the sums over finite subsets of E . Then every subset of X is Carathéodory measurable.

Indeed, for $A, E \subseteq X$, the index set E is the disjoint union of $E \cap A$ and $E \setminus A$. Additivity of nonnegative sums over a disjoint partition gives

$$\mu_w^*(E) = \mu_w^*(E \cap A) + \mu_w^*(E \setminus A).$$

Thus $\mathcal{M}_{\mu_w^*} = \mathcal{P}(X)$. The construction theorem may produce the entire power set, although it need not do so for a general outer measure.

5.2 Lebesgue measurable sets

Fix $d \geq 1$. Let $\mathcal{L}_d^*$ denote Lebesgue outer measure on $\mathbb{R}^d$, generated by countable covers by bounded open axis-parallel rectangles and their d -dimensional volumes. We next verify that the Carathéodory domain of $\mathcal{L}_d^*$ contains the Borel σ -algebra.

5.2.1 Open rectangles and Borel sets

The open rectangles used to define $\mathcal{L}_d^*$ are already Carathéodory measurable. The point is that the part of a covering rectangle outside a fixed open rectangle can be covered by finitely many open rectangles, with an arbitrarily small increase in total volume.

Proposition 5.2.1 (Bounded open rectangles are measurable). Every bounded open axis-parallel rectangle in $\mathbb{R}^d$ is Carathéodory measurable with respect to $\mathcal{L}_d^*$.

Proof. (proves 5.2.1)

Let R be a bounded open axis-parallel rectangle. Fix $E \subseteq \mathbb{R}^d$. Subadditivity gives

$$\mathcal{L}_d^*(E) \leq \mathcal{L}_d^*(E \cap R) + \mathcal{L}_d^*(E \cap R^c).$$

We prove the reverse inequality. It is automatic when $\mathcal{L}_d^*(E) = \infty$, so assume $\mathcal{L}_d^*(E) < \infty$, and fix $\varepsilon > 0$.

Step 1: choose an open-rectangle cover of E .

Choose bounded open axis-parallel rectangles $(Q_n)_{n \geq 1}$ that cover E and satisfy

$$\sum_{n=1}^{\infty} \text{vol}_d(Q_n) \leq \mathcal{L}_d^*(E) + \frac{\varepsilon}{2}.$$

For each n , the intersection

$$U_n = Q_n \cap R$$

is an open rectangle (or is empty), and the sets U_n cover $E \cap R$.

Step 2: cover the complementary part of each Q_n .

For every n , the set $Q_n \cap R^c$ has an open-rectangle cover $V_{n,1}, \dots, V_{n,m_n}$, where $m_n \leq 3^d$, such that

$$\sum_{r=1}^{m_n} \text{vol}_d(V_{n,r}) \leq \text{vol}_d(Q_n) - \text{vol}_d(U_n) + \varepsilon 2^{-n-1}.$$

The figure illustrates this elementary geometric fact in dimensions one and two.

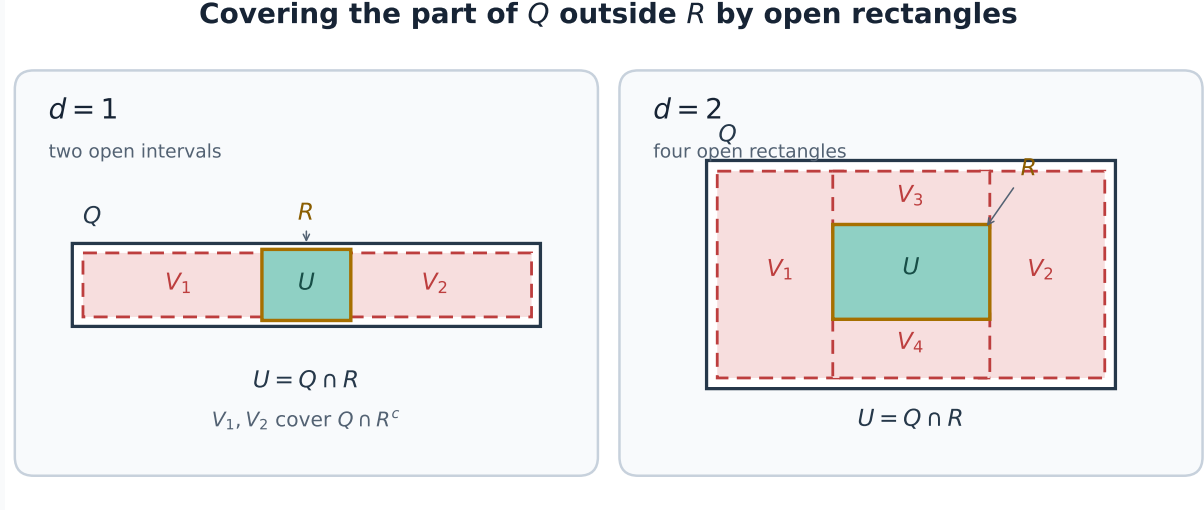


Figure 3: The part of an open rectangle Q outside an open rectangle R can be covered by at most 3^d open rectangles V_r , while $U = Q \cap R$ is itself open. Dashed boundaries indicate the arbitrarily small enlargement used to keep every covering rectangle open.

Figure description: Two diagrams. In dimension one, an interval Q contains an interval R ; their intersection U is the middle portion, and the two remaining portions are covered by slightly enlarged open intervals V one and V two. In dimension two, a rectangle Q contains a central rectangle R ; their intersection U is central, and the surrounding four regions are covered by slightly enlarged open rectangles V one through V four.

For $d = 1$, the complement of an open interval inside Q_n consists of its left and right portions. Enlarging each endpoint slightly gives two open intervals whose total length is within $\varepsilon 2^{-n-1}$ of the length of the complementary part. For $d = 2$, use the left and right strips of Q_n outside R , followed by the lower and upper strips in the middle; after slightly enlarging their boundaries, these are four open rectangles. Their total area can be made within $\varepsilon 2^{-n-1}$ of the area of the complementary part. In general, divide each coordinate direction into its left, middle, and right parts relative to R . The resulting grid has at most 3^d rectangular pieces, and every piece except the middle one lies outside R . Slightly enlarging those pieces gives the stated open-rectangle cover.

The rectangles $V_{n,r}$ cover $E \cap R^c$. Hence

$$\begin{aligned} \mathcal{L}_d^*(E \cap R) + \mathcal{L}_d^*(E \cap R^c) &\leq \sum_{n=1}^{\infty} \left(\text{vol}_d(U_n) + \sum_{r=1}^{m_n} \text{vol}_d(V_{n,r}) \right) \\ &\leq \sum_{n=1}^{\infty} \text{vol}_d(Q_n) + \sum_{n=1}^{\infty} \varepsilon 2^{-n-1} \\ &\leq \mathcal{L}_d^*(E) + \varepsilon. \end{aligned}$$

Letting $\varepsilon \downarrow 0$ proves the reverse inequality, so R is measurable. □

Corollary 5.2.2 (Borel sets are Carathéodory measurable). Every Borel subset of $\mathbb{R}^d$ is Carathéodory measurable with respect to $\mathcal{L}_d^*$.

Proof. (proves 5.2.2)

The bounded open rectangles with rational endpoints form a countable base for the Euclidean topology. Thus every open subset of $\mathbb{R}^d$ is a countable union of measurable open rectangles. Since the Carathéodory measurable sets form a σ -algebra, they contain the σ -algebra generated by the open sets, namely $\mathcal{B}(\mathbb{R}^d)$. $\square$

5.3 The coincidence criterion for measures

The Carathéodory theorem constructs a measure from an outer measure. The problem in this section is different: two measures are already defined on the same σ -algebra, and their values are known to agree only on a generating collection. Dynkin systems propagate that agreement to the whole σ -algebra.

Definition 5.3.1 (π -system and Dynkin system). Let X be a set.

π -system A collection $\mathcal{P} \subseteq \mathcal{P}(X)$ is a π -system if $A \cap B \in \mathcal{P}$ whenever $A, B \in \mathcal{P}$.

λ -system A collection $\mathcal{D} \subseteq \mathcal{P}(X)$ is a λ -system, if

full set $\emptyset, X \in \mathcal{D}$;

complement $A \in \mathcal{D}$ implies $A^c \in \mathcal{D}$

disjoint countable union $\bigcup_{n=1}^{\infty} A_n \in \mathcal{D}$ for every pairwise disjoint sequence $(A_n \in \mathcal{D})_{n \geq 1}$.

The smallest λ system containing a collection $\mathcal{C}$ is denoted by $\lambda(\mathcal{C})$.

The intersection of any family of λ systems is a λ system, and 2^X is a λ system. Thus $\lambda(\mathcal{C})$ exists and coincides with the intersection of all λ systems containing $\mathcal{C}$.

Lemma 5.3.2 (Difference closure in a λ system). If $\mathcal{D}$ is a λ system and $A, B \in \mathcal{D}$ satisfy $A \subseteq B$, then $B \setminus A \in \mathcal{D}$.

Proof. (proves 5.3.2)

The sets A and B^c are disjoint members of $\mathcal{D}$. After adjoining empty sets, closure under disjoint countable unions gives $A \cup B^c \in \mathcal{D}$. Closure under complements gives

$$B \setminus A = (A \cup B^c)^c \in \mathcal{D}.$$

$\square$

Theorem 5.3.3 (Dynkin's π - λ theorem). If $\mathcal{P}$ is a π -system on X , then

$$\lambda(\mathcal{P}) = \sigma(\mathcal{P}).$$

Proof. (proves 5.3.3)

The proof promotes intersection closure first from $\mathcal{P}$ to $\mathcal{P} \times \mathcal{D}$, and then from there to $\mathcal{D} \times \mathcal{D}$.

Step 1: reduce the theorem to intersection closure.

Set $\mathcal{D} = \lambda(\mathcal{P})$. Since every σ -algebra is a Dynkin system,

$$\mathcal{D} \subseteq \sigma(\mathcal{P}).$$

It remains to prove that $\mathcal{D}$ is a σ -algebra. For this, it is enough to prove that $\mathcal{D}$ is closed under finite intersections: such closure, together with closure under complements, gives closure under finite unions. If $(A_n)_{n \geq 1}$ is any sequence in $\mathcal{D}$, set

$$B_1 = A_1, \quad B_n = A_n \setminus \bigcup_{j=1}^{n-1} A_j \quad (n \geq 2).$$

Then $B_n \in \mathcal{D}$, the sets B_n are pairwise disjoint, and $\bigcup_{n \geq 1} A_n = \bigcup_{n \geq 1} B_n$. Closure under disjoint countable unions therefore gives $\bigcup_{n \geq 1} A_n \in \mathcal{D}$, proving that $\mathcal{D}$ is a σ -algebra once finite-intersection closure is established.

Step 2: intersect an arbitrary Dynkin member with a generator.

Fix $A \in \mathcal{P}$ and define

$$\mathcal{D}_A = \{B \in \mathcal{D} : A \cap B \in \mathcal{D}\}.$$

This is a Dynkin system. Indeed, $A \cap X = A \in \mathcal{D}$. If $B \in \mathcal{D}_A$, then $A \cap B \subseteq A$, and [Lemma 5.3.2](#) gives

$$A \cap B^c = A \setminus (A \cap B) \in \mathcal{D}.$$

If the sets $B_n \in \mathcal{D}_A$ are pairwise disjoint, then the sets $A \cap B_n$ are pairwise disjoint members of $\mathcal{D}$, and hence

$$A \cap \bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} (A \cap B_n) \in \mathcal{D}.$$

Thus $\mathcal{D}_A$ is a Dynkin system. Since $\mathcal{P}$ is a π -system, $A \cap B \in \mathcal{P} \subseteq \mathcal{D}$ for every $B \in \mathcal{P}$. Therefore $\mathcal{P} \subseteq \mathcal{D}_A$, and minimality of $\mathcal{D}$ gives $\mathcal{D}_A = \mathcal{D}$. We have proved

$$A \cap B \in \mathcal{D} \quad \text{whenever } A \in \mathcal{P} \text{ and } B \in \mathcal{D}. \quad (2)$$

Step 3: extend intersection closure to two Dynkin members.

Now fix $B \in \mathcal{D}$ and define

$$\mathcal{E}_B = \{A \in \mathcal{D} : A \cap B \in \mathcal{D}\}.$$

The same complement, proper-difference, and disjoint-union argument shows that $\mathcal{E}_B$ is a Dynkin system. By (2), it contains $\mathcal{P}$. Hence $\mathcal{E}_B = \mathcal{D}$. Since $B \in \mathcal{D}$ was arbitrary, $\mathcal{D}$ is closed under finite intersections. Thus it is a σ -algebra, so $\sigma(\mathcal{P}) \subseteq \mathcal{D}$. $\square$

5.3.1 Finite and sigma-finite measures

Theorem 5.3.4 (Coincidence criterion for finite measures). Let $\mathcal{P}$ be a π -system on X , and let $\mathcal{A} = \sigma(\mathcal{P})$. Suppose μ and ν are finite measures on $(X, \mathcal{A})$ such that

$$\mu(X) = \nu(X) \quad \text{and} \quad \mu(P) = \nu(P) \quad \text{for every } P \in \mathcal{P}.$$

Then $\mu(A) = \nu(A)$ for every $A \in \mathcal{A}$.

Proof. (proves 5.3.4)

Consider the coincidence class

$$\mathcal{D} = \{A \in \mathcal{A} : \mu(A) = \nu(A)\}.$$

The equality of the total masses gives $X \in \mathcal{D}$. If $A \in \mathcal{D}$, finiteness gives

$$\mu(A^c) = \mu(X) - \mu(A) = \nu(X) - \nu(A) = \nu(A^c),$$

so $A^c \in \mathcal{D}$. If $(A_n \in \mathcal{D})_{n \geq 1}$ is pairwise disjoint, countable additivity gives

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n) = \sum_{n=1}^{\infty} \nu(A_n) = \nu\left(\bigcup_{n=1}^{\infty} A_n\right).$$

Thus $\mathcal{D}$ is a Dynkin system. It contains $\mathcal{P}$, so Theorem 5.3.3 gives $\mathcal{A} = \sigma(\mathcal{P}) \subseteq \mathcal{D}$. □

Remark (When the generator contains the whole space). If $X \in \mathcal{P}$, agreement on $\mathcal{P}$ already includes the hypothesis $\mu(X) = \nu(X)$. The separate total-mass hypothesis is needed when $X \notin \mathcal{P}$.

For measures with infinite total mass, apply the finite result on an increasing exhaustion.

Theorem 5.3.5 (Coincidence criterion for sigma-finite measures). Let $\mathcal{P}$ be a π -system on X , let $\mathcal{A} = \sigma(\mathcal{P})$, and let μ, ν be measures on $(X, \mathcal{A})$. Assume that:

1. $\mu(P) = \nu(P)$ for every $P \in \mathcal{P}$;
2. there is an increasing sequence $(P_n \in \mathcal{P})_{n \geq 1}$ such that

$$X = \bigcup_{n=1}^{\infty} P_n \quad \text{and} \quad \mu(P_n) = \nu(P_n) < \infty \quad \text{for every } n.$$

Then $\mu(A) = \nu(A)$ for every $A \in \mathcal{A}$.

Proof. (proves 5.3.5)

Exercise □

Warning (Finiteness cannot be omitted). Agreement on a generating π -system does not by itself imply agreement on the generated σ -algebra. On $\mathbb{N}$, let

$$T_n = \{n, n+1, n+2, \dots\}, \quad \mathcal{P} = \{T_n : n \in \mathbb{N}\}.$$

The family $\mathcal{P}$ is a π -system and generates $\mathcal{P}(\mathbb{N})$, since $\{n\} = T_n \setminus T_{n+1}$ and every subset of $\mathbb{N}$ is a countable union of singletons. Counting measure μ and the measure

$$\nu(E) = \begin{cases} 0, & E = \emptyset, \\ \infty, & E \neq \emptyset \end{cases}$$

are countably additive: for ν , both sides of the countable-additivity identity are 0 when every set is empty and are ∞ otherwise. They agree on every T_n , but $\mu(\{n\}) = 1 \neq \infty = \nu(\{n\})$. The measure ν has no countable exhaustion by sets of finite ν -measure.

5.3.2 Application to Borel measures

Theorem 5.3.6 (Coincidence on half-open rectangles). Let μ and ν be Borel measures on $\mathbb{R}^d$ that are finite on compact sets. Suppose

$$\mu \left(\prod_{j=1}^d (a_j, b_j] \right) = \nu \left(\prod_{j=1}^d (a_j, b_j] \right)$$

whenever $-\infty < a_j < b_j < \infty$ for every j . Then $\mu = \nu$ on $\mathcal{B}(\mathbb{R}^d)$.

Proof. (proves 5.3.6)

The proof verifies the three inputs of the σ -finite coincidence criterion: a π -system, generation, and a finite exhaustion.

Step 1: verify the π -system property.

Let $\mathcal{P}$ consist of $\emptyset$ and all bounded half-open rectangles $\prod_{j=1}^d (a_j, b_j]$. Intersections of such rectangles are again half-open rectangles or empty, so $\mathcal{P}$ is a π -system.

Step 2: show that the half-open rectangles generate the Borel sets.

Every member of $\mathcal{P}$ is Borel. Conversely, if $R = \prod_{j=1}^d (a_j, b_j]$ is an open rectangle, then R is the union of all half-open rectangles

$$\prod_{j=1}^d (p_j, q_j] \quad \text{with rational endpoints and} \quad a_j < p_j < q_j < b_j.$$

This is a countable union. Thus open rectangles, and hence all open sets, belong to $\sigma(\mathcal{P})$. Therefore $\sigma(\mathcal{P}) = \mathcal{B}(\mathbb{R}^d)$.

Step 3: construct a common finite exhaustion.

The rectangles

$$P_n = (-n, n]^d, \quad n \geq 1,$$

form an increasing sequence in $\mathcal{P}$ whose union is $\mathbb{R}^d$. Each P_n is contained in the compact cube $[-n, n]^d$. Finiteness on compact sets and monotonicity give

$$\mu(P_n) = \nu(P_n) < \infty.$$

The result now follows from [Theorem 5.3.5](#). □

Corollary 5.3.7 (Uniqueness of Lebesgue measure). Lebesgue measure on the Borel subsets of $\mathbb{R}^d$ is the unique Borel measure that is finite on compact sets and assigns every bounded half-open rectangle its d -dimensional volume. Its completion is the measure λ_d on $\mathcal{L}(\mathbb{R}^d)$.